



POLITECNICO
MILANO 1863

Experimental Dynamics and Data Processing

FRF – random

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Problem: **people** can be considered and modelled as mechanical systems therefore it is reasonable to think that they **interact with the mechanical/civil structures** on which they stand

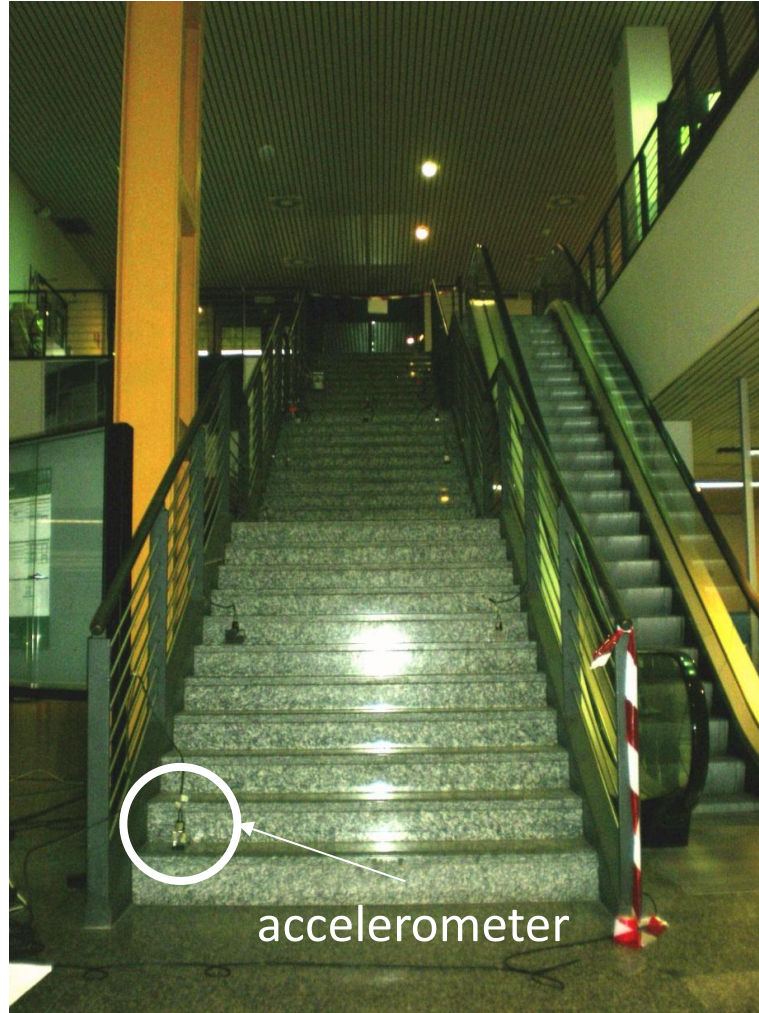
Goal: to **investigate** the effect of the human-structure interaction by looking at **how the dynamical characteristics** of the structure **change** when humans are on it.



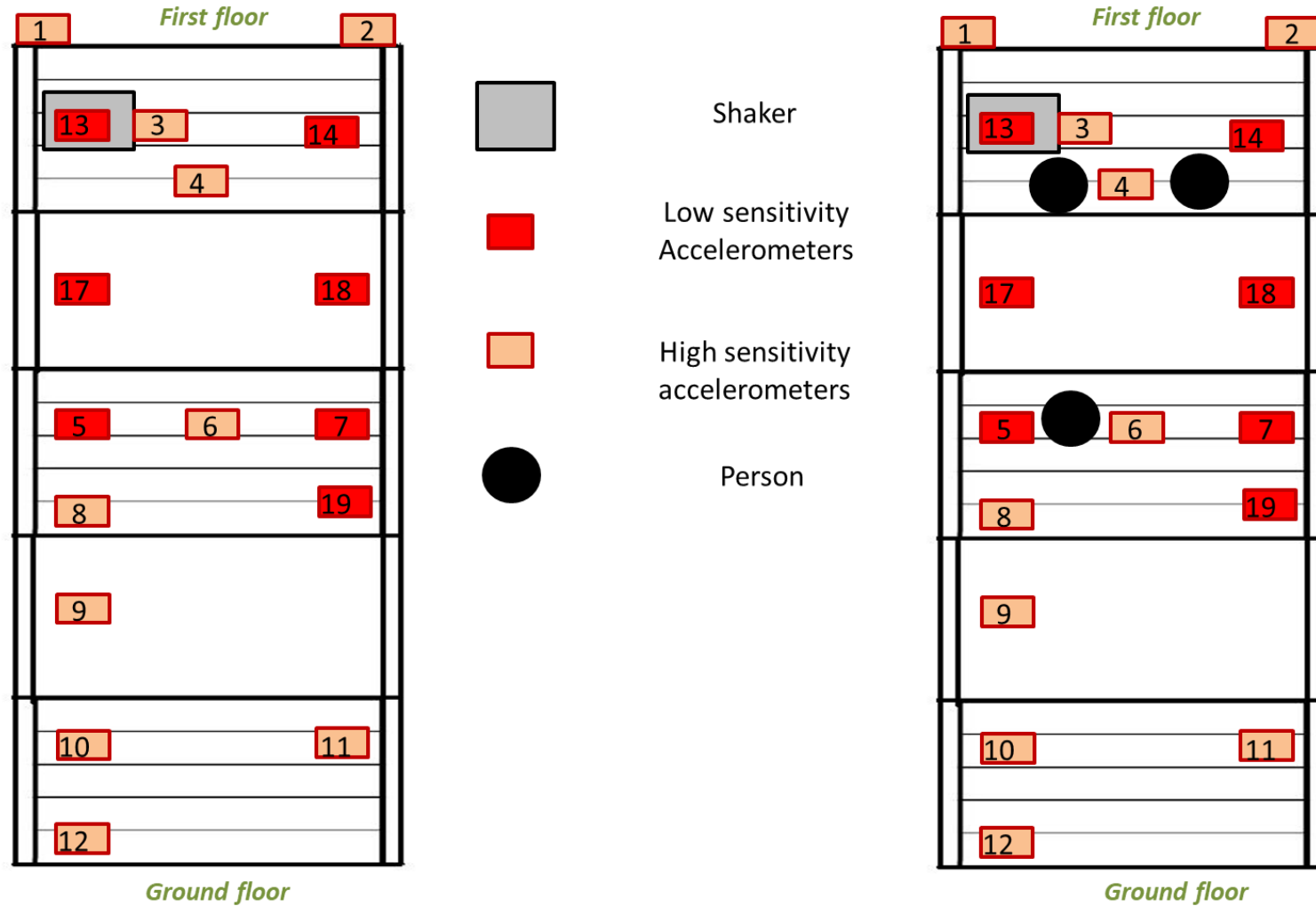
The response of the structure to a known input should be studied when people are or not on the structure itself.

As a first step the **cross-spectrum** between the input and the output of the system has to be analysed.

Test structure: Staircase of B12 Building in Bovisa Campus (Milano)



Measurement set-up



Problem: the spectrum of random signals cannot be easily interpreted as it is



It is possible to perform an **averaging** process on several sub-records of the original signal.

Power spectrum:

$$\begin{aligned} S_{AA}(f_k) &= E[\hat{A}_j^*(f_k) \hat{A}_j(f_k)] = \\ &= \lim_{n_d \rightarrow \infty} \frac{1}{n_d} \sum_{i=1}^{n_d} \hat{A}_i^*(f_k) \hat{A}_i(f_k) \end{aligned}$$

Which are the **main information** related to the cross-spectrum?

The cross-spectrum indicates the **correlation level** of two signals in a given frequency range

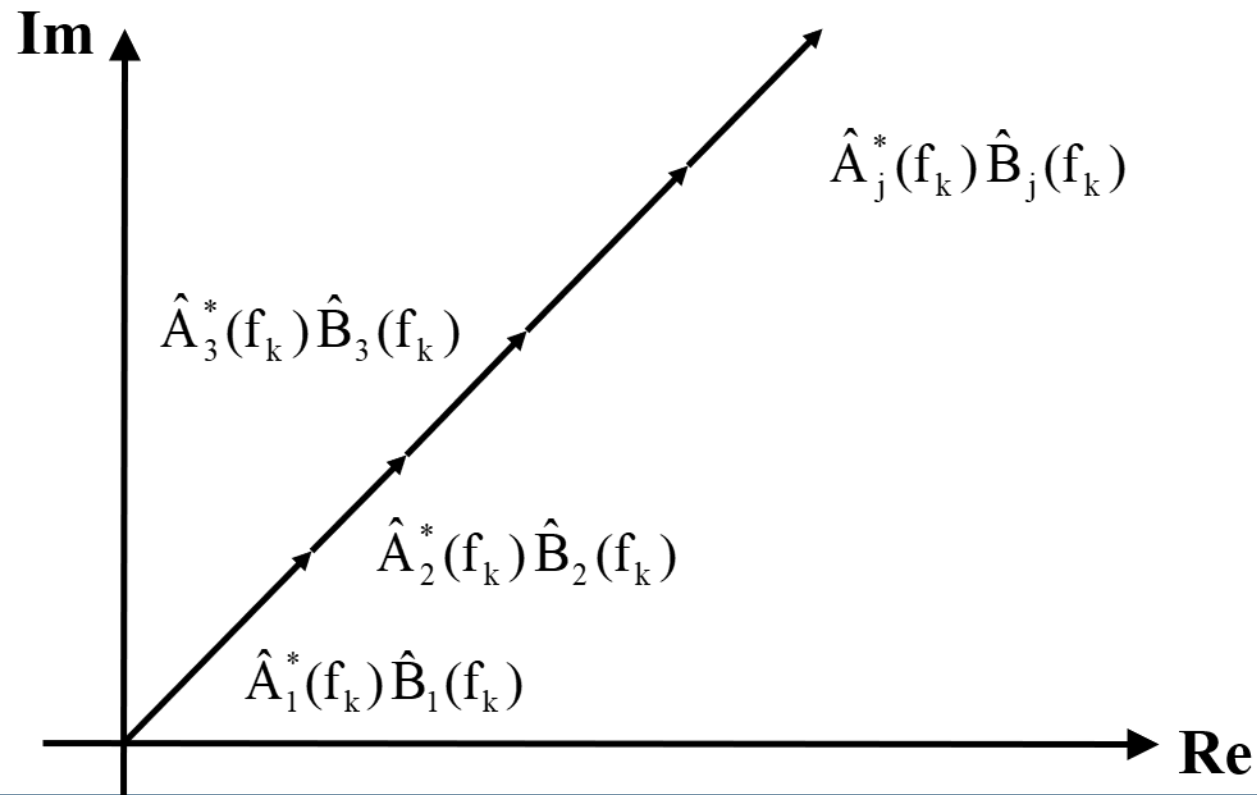
$$G_{xy}(f_k) = \frac{1}{N} * \sum_{i=1}^N (\text{conj}(X(f_k)) * Y(f_k))$$

The **average process** allows to increase statistical reliability of the result

Effect of the averaging process on the cross-spectrum evaluation

Complete correlation between the two signals:

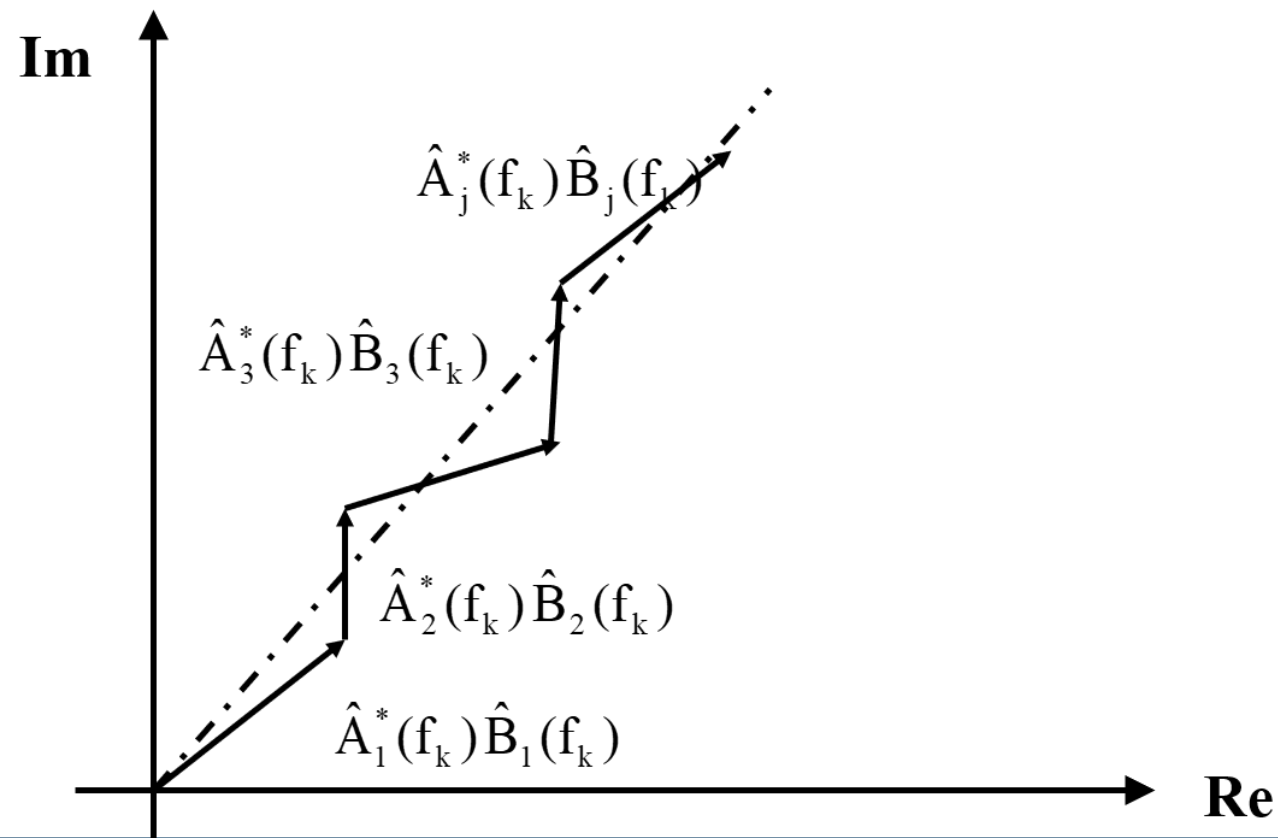
The **phase shift** between the signals **doesn't change** with varying the considered time history



Effect of the averaging process on the cross-spectrum evaluation

Partial correlation between the two signals:

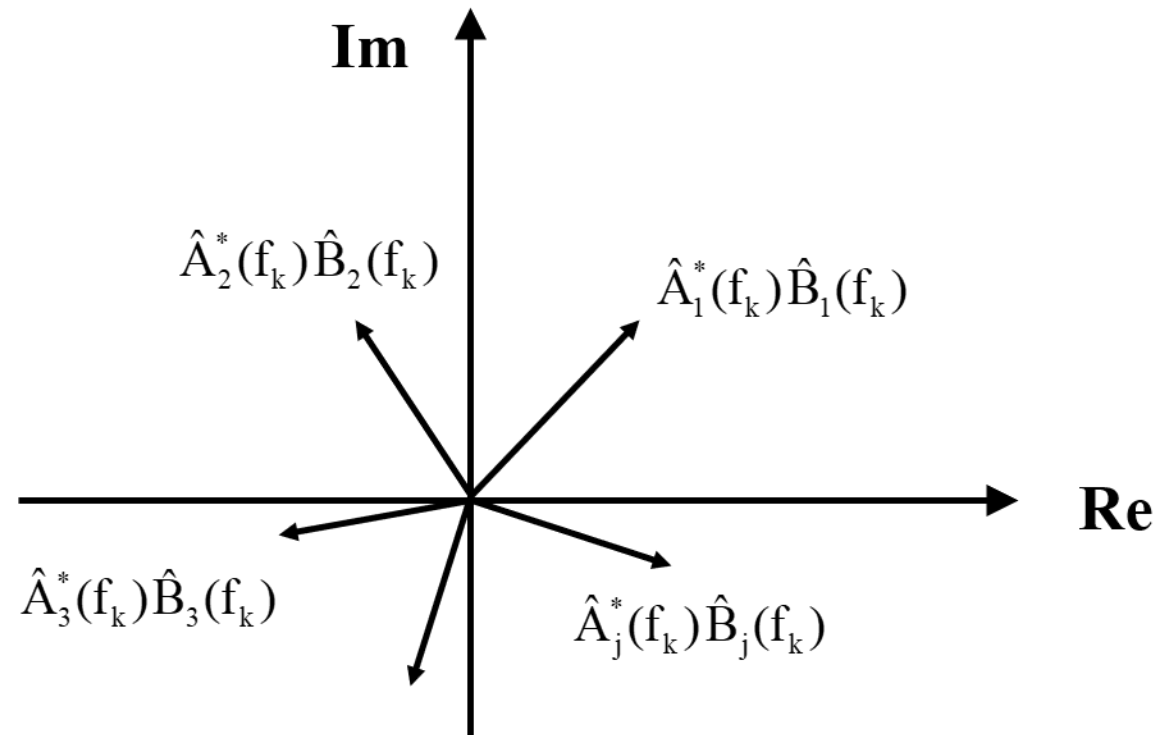
The **phase shift** between the signals **change** with varying the considered time history



Effect of the average process on the cross-spectrum evaluation

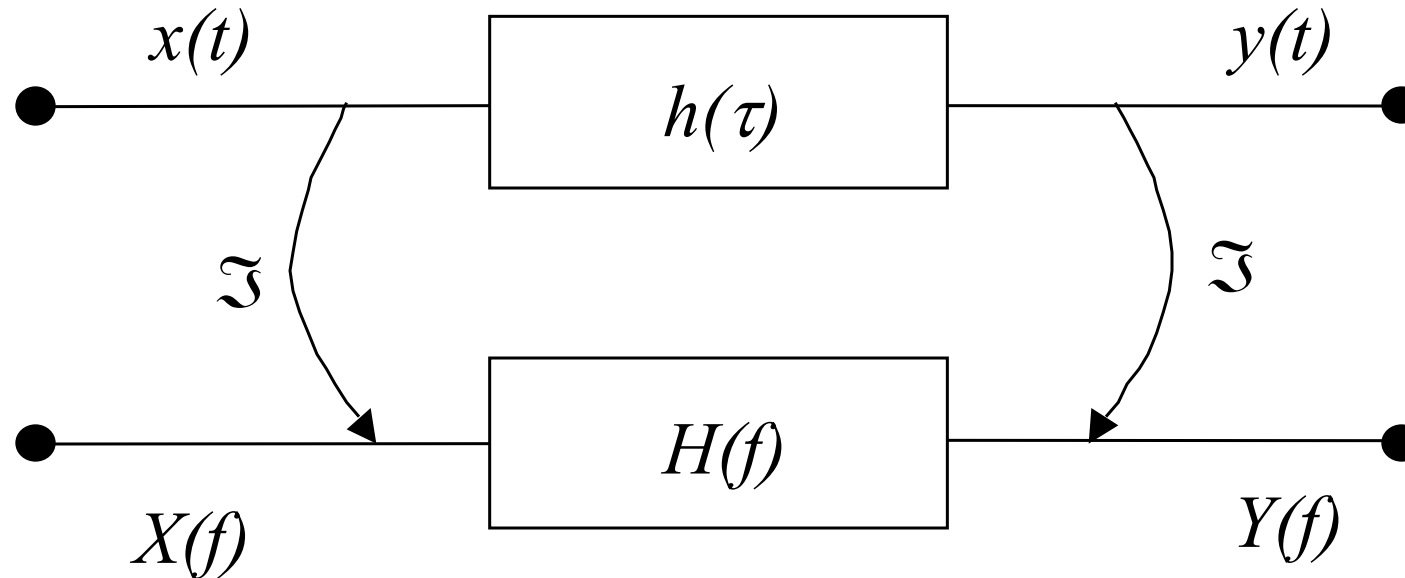
Lack of correlation between the two signals:

The **phase shift changes randomly** with varying the considered time history ($n_d \rightarrow \infty$ $S_{AB} \rightarrow 0$)



Transfer Functions

For any system, **the transfer function** is defined as



$$y(t) = \int_0^{\infty} h(\tau) x(t - \tau) d\tau$$

Convolution in the time domain



Product in the frequency domain

$$Y(f) = X(f) \cdot H(f)$$

Frequency Response Function of
the system

$$Y(f) = X(f) \cdot H(f)$$

$$X^*(f) Y(f) = H(f) X^*(f) X(f)$$

$$G_{xy}(f) = H(f) G_{xx}(f)$$

$$H(f) = \frac{Y(f)}{X(f)} \equiv \frac{G_{xy}(f)}{G_{xx}(f)} = H_1(f)$$

H1 estimator

- is not affected by the uncorrelated noise on the output measurement;
- underestimate the frequency response function in the case of noise on the input measurement

$$Y(f) = X(f) \cdot H(f)$$

$$Y^*(f) Y(f) = H(f) Y^*(f) X(f)$$

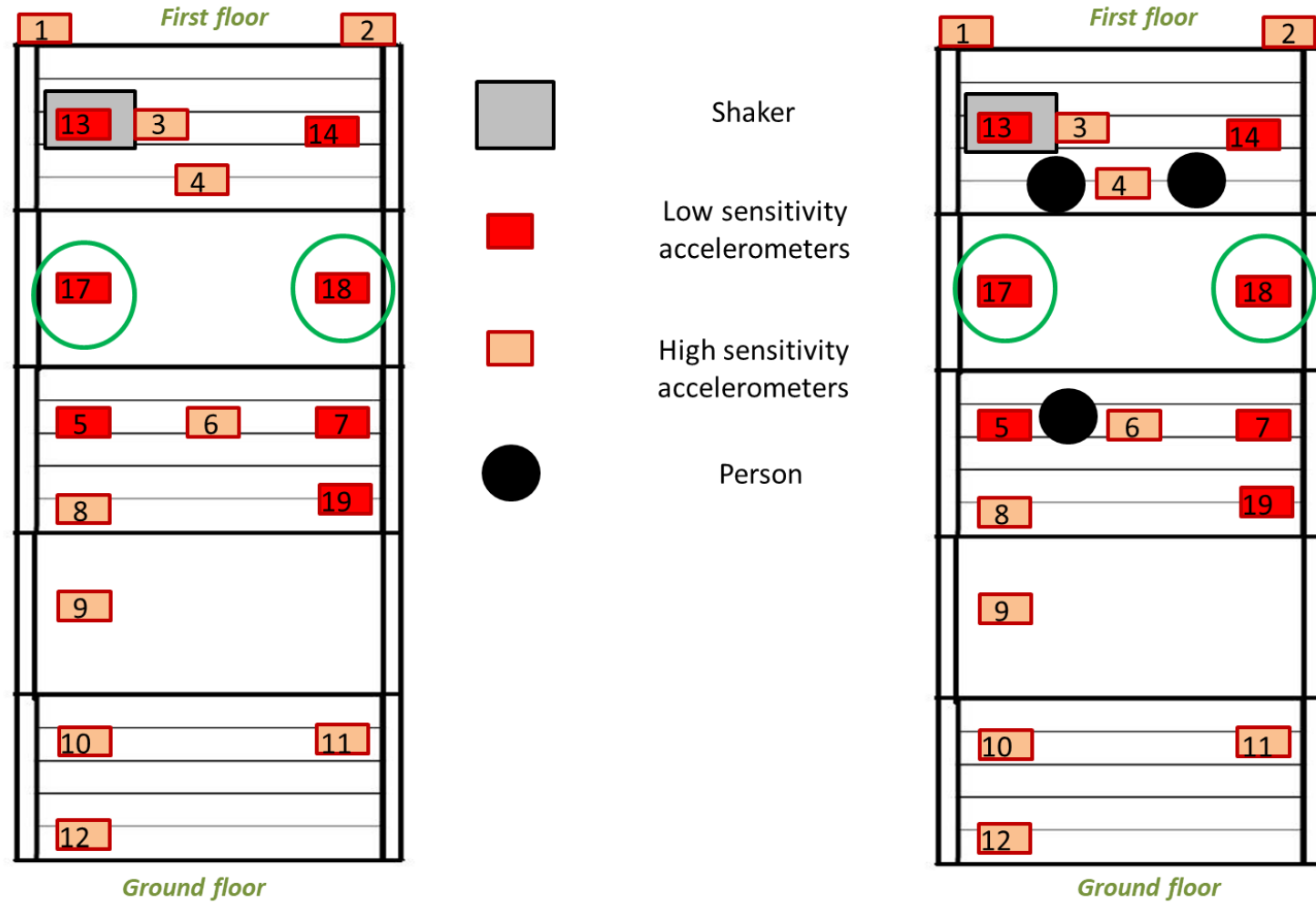
$$G_{yy}(f) = H(f) G_{yx}(f)$$

$$H(f) = \frac{Y(f)}{X(f)} \equiv \frac{G_{yy}(f)}{G_{yx}(f)} = H_2(f)$$

H2 estimator

- Is not affected by the uncorrelated noise on the input measurement;
- Overestimates the transfer function when there is noise on the output measurement

Test set-up



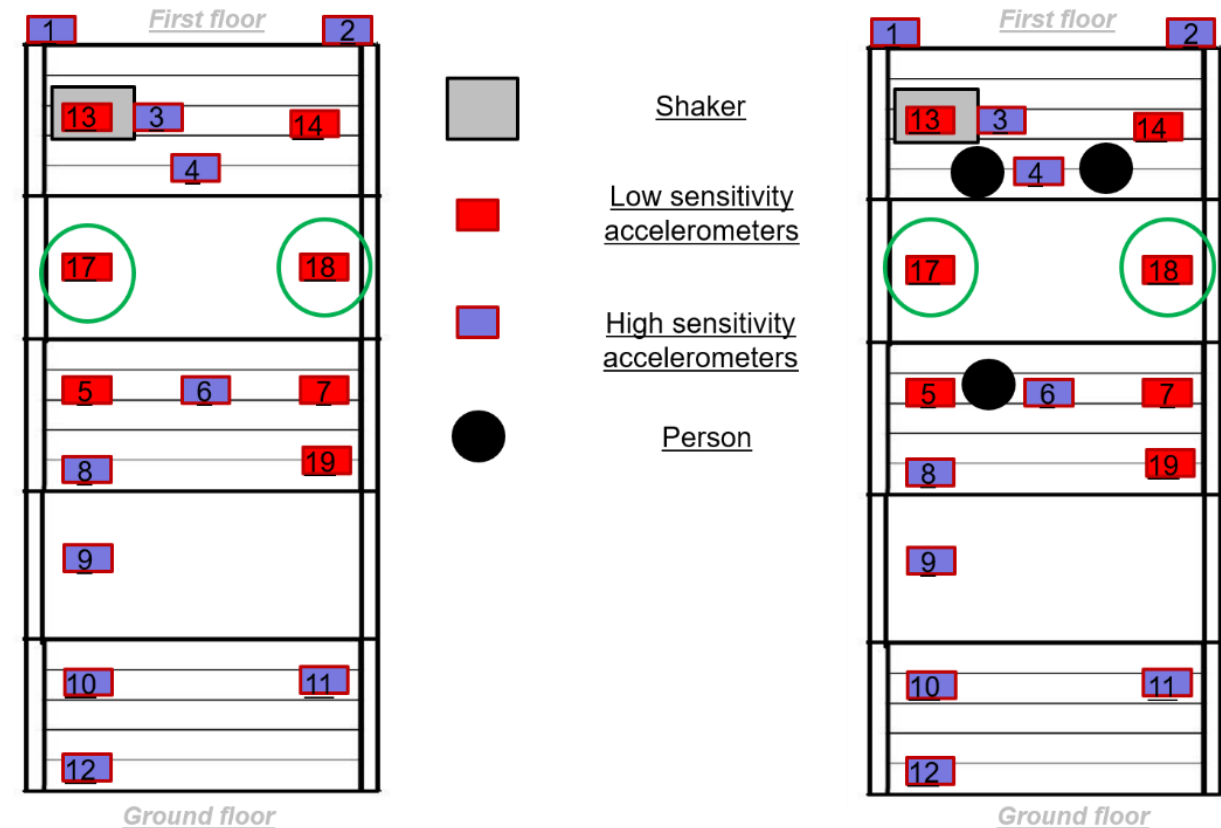
To do list

The data sets are related to 2 different tests on the staircase:

- «Lab_04_staircase_empty»: no people on the staircase
- «Lab_04_staircase_people»: 3 people on the staircase

Each file contains the sampling frequency (fsamp) and the time histories of:

- The forcing term (random input) in $\text{Data}(:,1)$
- Acceleration measured at point 17 in $\text{Data}(:,2)$
- Acceleration measured at point 18 in $\text{Data}(:,3)$



1. Evaluate, for both the cases (with and without people):
 - ✓ Power-spectra of the input and the output (both accelerometer 17 and 18), averaged using time histories of 30 s and 60 s
 - ✓ Cross-spectrum averaged using time histories of 30 s and 60 s.

Is the structure response correlated to the input?

Is the information given by the **cross-spectrum enough** to answer the previous question?

Otherwise, **what should we look at?**

2. Evaluate the Frequency response function between the input and the output as the Ratio of the Fourier transforms (**H**=Y/X) of the output and input

3. Evaluate the Frequency response function between the input and the output as **H1**, using time windows of 30 s and 60 s
4. Evaluate the Frequency response function between the input and the output as **H2**, using time windows of 30 s and 60 s

Which is the **correct way** to represent the actual FRF of the system?

Is it possible to identify which one of the proposed methods to calculate the FRF is the best one to represent the actual FRF of the system under analysis?

Plot of the spectrum

The spectrum is a **complex quantity**

It can be represented in different ways:

- Modulus and phase
- Real and imaginary part
- Complex plane Re-Im

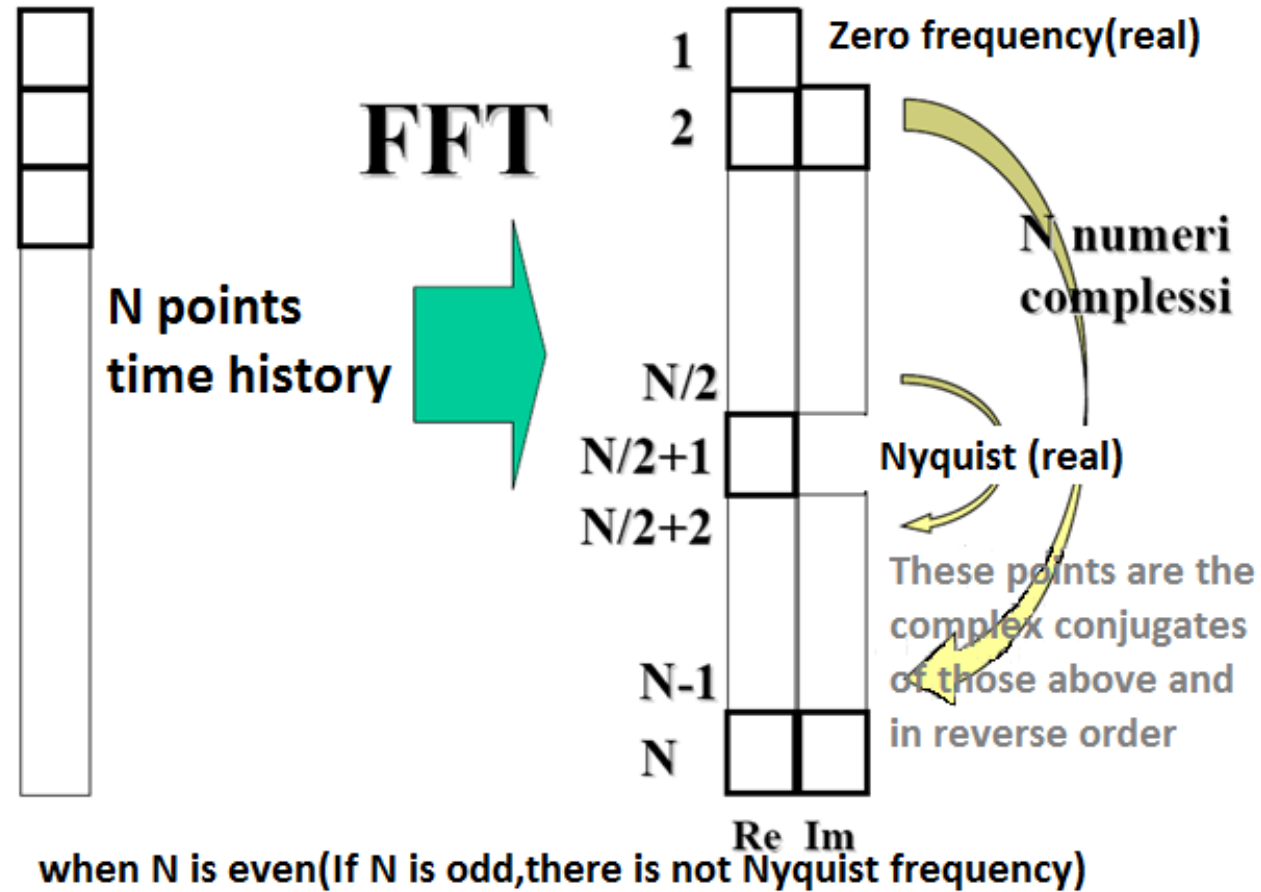


As function of frequency

Discrete fourier transform

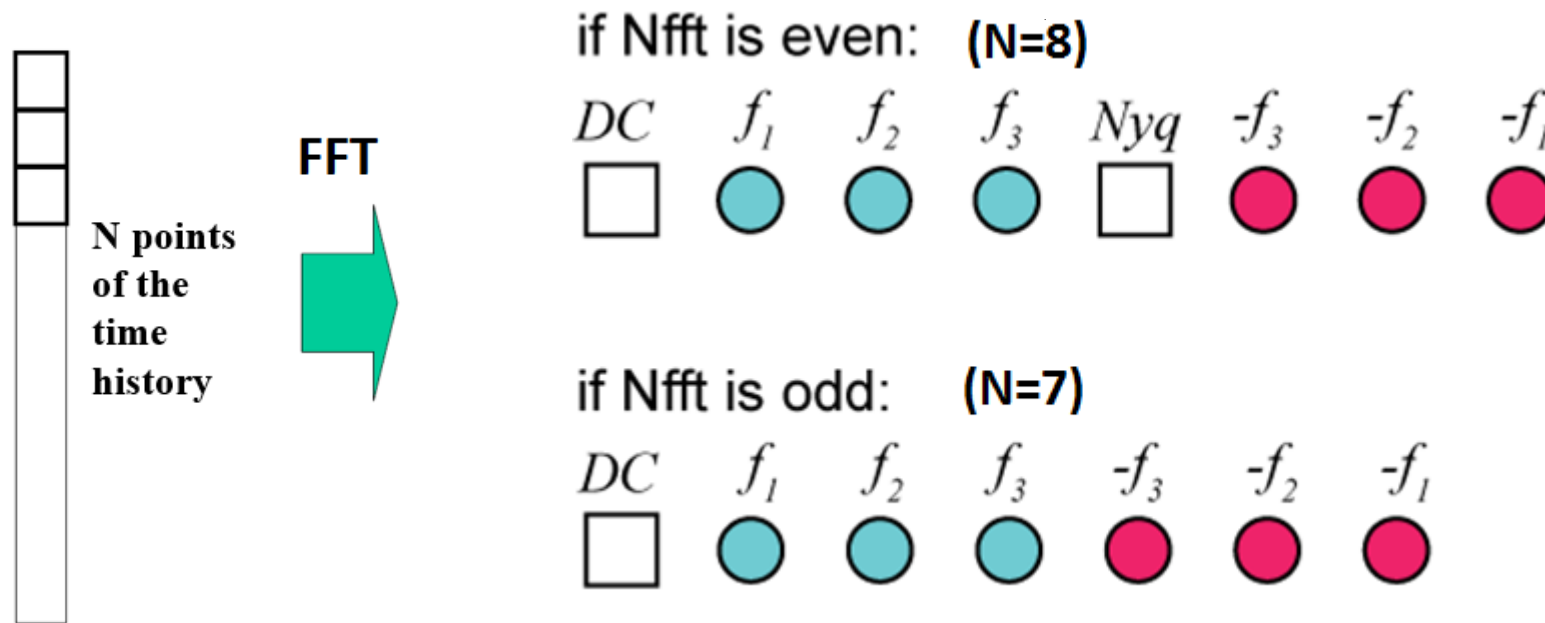
FFT Matlab

Pay attention:



Discrete fourier transform: FFT Matlab

Pay attention:



Results from a MATLAB fft. Each symbol represents an element in the vector, squares are unique (DC and Nyquist if present), circles are positive (blue) and negative (red) frequencies

Discrete fourier transform: FFT Matlab

Pay attention: The FFT function includes a scaling factor N between the original and transformed signals. To estimate the correct amplitude of spectral components, the FFT output must be rescaled by dividing by N .

It is possible to consider just the positive frequencies but...

N even $\rightarrow$
consider $(N/2+1)$ points
 $f_{\max} = \text{Nyquist}$

N odd $\rightarrow$
consider $((N+1)/2)$ points
 $f_{\max} = \text{Nyquist-df}/2$

We need to normalize in the right way

$y(1)=y(1)/N$
 $y(2:N/2)= y(2:N/2)*2/N$
 $y(N/2+1)=y(N/2+1)/N$

$y(1)=y(1)/N$
 $y(2:(N+1)/2)= y(2:(N+1)/2)*2/N$

Fourier Transform: the Leakage error

Problem

For a given signal it is **not always possible** to identify a period and therefore sample it for an integer number of periods



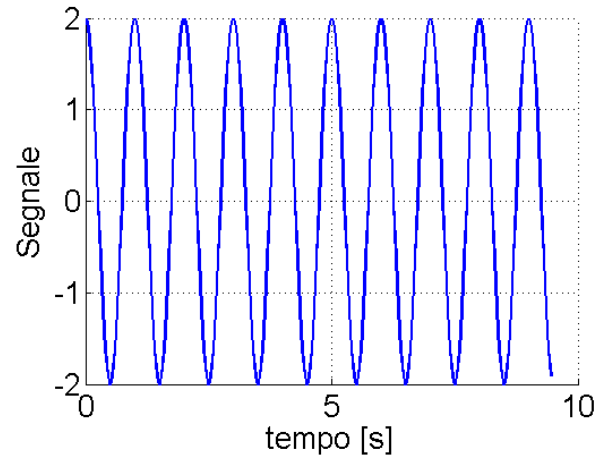
Leakage



Problem mitigation

It is possible to **use windows** different from the rectangular one. Each window modifies the signal and its spectrum in a different way.

The choice of the best window depends on the signal under analysis and on the goals of the measurement



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